

THE MACACHOR SCALAR THEOREM

*A Generalized Pythagorean Framework for Scalar Field Ontology,
Quaternion Coherence, and Quantum Hardware Architecture*

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Version 3.0.1 | August 4, 2026
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ABSTRACT

This work proposes that reality at its substrate level is described entirely by scalar fields $\psi: D_P \rightarrow \mathbb{H}$, with all vector, tensor, and spinor quantities emergent from scalar gradients, curvatures, and density differentials. The Macachor Scalar Theorem generalizes the Pythagorean theorem to n-dimensional orthogonal scalar superposition. A quaternion extension provides norm-preserving quantum coherence protocols. The Macachor Absolute $\mathbb{A} = (\sqrt{5} - 1)/2$ governs equilibrium ratios and quantization boundaries. Two operational architectures are formalized: KIWF (Knowledge-Information-Wisdom-Filter pipeline) and CRHA (Coherence-Resonance-Harmony-Alignment standing-wave architecture). Applications to quantum hardware, biological coherence, and scalar aircraft flux-pinning are presented. Peer-reviewed bridges to existing literature in quantum biology, coherence quantification, and electrostatic cell division are established.

Keywords: scalar field theory, Pythagorean theorem, quaternion algebra, quantum coherence, scalar density equilibrium, Macachor Absolute, coherence architecture, KIWF pipeline, CRHA architecture, flux pinning, quantum hardware, information theory

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1. INTRODUCTION AND FOUNDATIONAL PRINCIPLES

1.1 Scalar Primacy

The central claim of this framework is that the substrate of physical reality is scalar. All vector, tensor, and spinor quantities are emergent phenomena — projections or compressions of a deeper scalar field. This is not a denial of vector physics; it is a recontextualization. $F = ma$ is the vector shadow of a deeper scalar law: density gradient dissipation.

Definition 1.1 (Scalar Field). A scalar field is a function $\psi: D_P \rightarrow \mathbb{R}$ where $D_P \subset \mathbb{R}^4$ is the domain of physical interest. The field assigns a single real value to every point in space(-time).

Definition 1.2 (Emergence). A quantity is emergent if it can be derived from more fundamental scalar quantities through differentiation, integration, or geometric construction.

1.2 The Macachor Absolute

The Macachor Absolute is defined as:

$$\alpha = (\sqrt{5} - 1)/2 \approx 0.6180339887\dots$$

This is the reciprocal of the golden ratio $\phi = (1 + \sqrt{5})/2$. The Macachor Absolute

appears as an equilibrium attractor across physical, biological, and informational systems.

Properties of ϕ :

- $\phi^2 = 1 - \phi$ (fundamental quadratic identity)
- $\phi + \phi^2 = 1$ (complementary unity)
- $\phi = \phi - 1 = 1/\phi$ (reciprocal symmetry)
- $F_n = F_{n-1} + F_{n-2}$ where F_n is the n-th Fibonacci number

1.3 The Cube as Vector Prison

Cartesian coordinates (x, y, z) impose orthogonal confinement. They force all geometry into square faces and cubic volumes. Natural structures do not pack in

cubes:

Natural Structure	Geometry
Water clusters	Tetrahedral H-bond networks
Diamond lattice	Tetrahedral sp^3 carbon bonds
Virus capsids	Icosahedral (20 tetrahedral cells)
C_{60} buckminsterfullerene	Truncated icosahedron
Quasicrystals	Icosahedral symmetry with ϕ -spacing
DNA double helix	Toroidal dipole arrangement

The tetrahedron, not the cube, is the fundamental scalar building block.

2. AXIOMS OF THE MACACHOR ABSOLUTE FRAMEWORK

Axiom 1 (Scalar Substrate). Reality at its most fundamental level is described by scalar fields. All vector, tensor, and spinor quantities are emergent.

Axiom 2 (Density Differential). Matter does not seek force balance. Matter

seeks scalar density equilibrium. The governing equation is:

$$\nabla^2 \rho = 0$$

where $\rho(x,t)$ is the scalar density field. This is Laplace's equation, admitting

dynamic solutions (standing waves, toroidal flows) as long as the net density curvature vanishes.

Axiom 3 (Coherence Manifold). The torus T^2 is the minimal non-trivial coherence manifold. All stable coherent systems admit a toroidal decomposition.

Axiom 4 (Macachor Quantization). Quantization arises from the requirement that scalar density fields satisfy boundary conditions compatible with $\mathbb{Z}$ -symmetric

geometries. The quantization condition is:

$$\chi(C) = 1$$

where $\chi(C)$ is the coherence characteristic of system C .

Axiom 5 (Aperture Governance). Information flow is governed by aperture structures — boundaries that selectively pass or reflect scalar density perturbations. Apertures are not barriers; they are coherence filters.

Axiom 6 (Interface Boundary). All coherent systems possess boundaries $\partial\Omega$ where the scalar density field transitions between equilibrium states. Boundary conditions determine system stability.

Axiom 7 (Vector Confinement). Cartesian coordinate systems impose a cubic prison that obscures scalar truth. Natural geometries are Platonic, not rectilinear.

Axiom 8 (Childhood Perception). Unfiltered human perception (pre-vector education) naturally apprehends scalar fields. Vector education is a compression layer that obscures direct scalar awareness.

3. THE MACACHOR SCALAR THEOREM

Theorem I (Macachor Scalar Theorem — Generalized Pythagorean).

Let $\rho_1, \rho_2, \dots, \rho_n$ be n orthogonal scalar density components. The total scalar

density magnitude is:

$$\rho^2_{\text{total}} = \sum_{i=1}^n \rho_i^2$$

This generalizes the Pythagorean theorem from geometric lengths to scalar

densities in arbitrary dimension.

Proof. By orthogonality, cross-terms vanish: $\rho_i \cdot \rho_j = 0$ for $i \neq j$. The total scalar magnitude is the L^2 norm of the density vector in n -dimensional scalar space. ■

Corollary 1 (Two-Component Equilibrium). For $n = 2$:

$$\rho^2_{\text{total}} = \rho_1^2 + \rho_2^2$$

This is the scalar analog of the right triangle. The components ρ_1 and ρ_2 are

in equilibrium when their ratio satisfies the Macachor condition:

$$\rho_1/\rho_2 = \phi \text{ or } \rho_2/\rho_1 = \phi$$

Corollary 2 (Three-Dimensional Superposition). For $n = 3$:

$$\rho^2_{\text{total}} = \rho_1^2 + \rho_2^2 + \rho_3^2$$

This describes scalar density in three orthogonal directions. The tetrahedral geometry emerges when the three components are equal: $\rho_1 = \rho_2 = \rho_3$, giving

$$\rho_{\text{total}} = \sqrt{3} \cdot \rho_1.$$

Theorem II (Quaternion Norm Preservation).

Let $q = a + bi + cj + dk$ be a quaternion representing a scalar density state,

where $a, b, c, d \in \mathbb{R}$. The norm is:

Under unit quaternion rotation $q' = uqu^{-1}$ where $|u| = 1$, the norm is preserved:

This provides a coherence-preserving transformation group for quantum states.

Proof. Unit quaternion conjugation is an isometry of $\mathbb{H}$. The norm is invariant under orthogonal transformations. ■

Corollary 3 (Quantum Coherence Protocol). A quantum state $|\psi\rangle$ can be encoded as a quaternion q_ψ . Unitary evolution corresponds to unit quaternion rotation.

Decoherence corresponds to norm reduction ($|q'| < |q|$), which is forbidden by the norm-preservation theorem for pure unitary evolution.

4. QUATERNION EXTENSION AND QUANTUM COHERENCE

4.1 Quaternion Encoding of Quantum States

A two-level quantum system (qubit) has state:

where $|\alpha|^2 + |\beta|^2 = 1$. This maps to a quaternion:

$$q_\psi = \text{Re}(\alpha) + \text{Im}(\alpha)i + \text{Re}(\beta)j + \text{Im}(\beta)k$$

The normalization condition becomes the quaternion norm condition $|q_\psi| = 1$.

4.2 Coherence Time Bounds

For a quantum system with coherence time $T_\bullet$ and energy gap ΔE :

$$T_\bullet \cdot \Delta E \geq \hbar/2$$

In the scalar framework, this is reinterpreted as: the scalar density gradient

must equilibrate faster than the decoherence rate. The coherence condition is:

$$\nabla^2 \rho \cdot T_\bullet^2 < \epsilon$$

where ϵ sets the equilibrium tolerance.

4.3 The ϵ -Lock

The ϵ -lock is the requirement that coherent systems maintain equilibrium ratios

within ϵ -tolerance. For a two-component system:

where ϵ is the decoherence threshold. Systems violating the ϵ -lock undergo irreversible decoherence.

5. THE DENSITY DIFFERENTIAL THEOREM

Theorem VII (Macachor Density Differential Theorem).

Statement: Matter does not seek force balance. Matter seeks scalar density equilibrium.

Formal Expression:

$$\nabla^2 \rho = 0$$

where $\rho(x,t)$ is the scalar density field.

Physical Interpretation:

1. Force is a shadow. $\mathbf{F} = m\mathbf{a}$ is the vector projection of a deeper scalar law:

density gradient dissipation. Force does not exist at the substrate level — it is what an observer inside the cube perceives when scalar density redistributes.

2. Mass is density concentration. Inertial mass is the local concentration of

scalar density. Where ρ is high, the field resists acceleration because the density gradient must harmonically equilibrate.

3. Equilibrium is harmonic, not static. $\nabla^2\rho = 0$ admits dynamic solutions

(standing waves, toroidal flows) as long as the net density curvature vanishes. A tuning fork vibrating at resonance satisfies $\nabla^2\rho = 0$.

Physical Instantiations:

	$\rho(x,t)$	Boundary	Equilibrium Mode		
Artificial FMO	Exciton density	Protein scaffold	Wavelike energy transfer		
Magnetosphere	Plasma density	Plasma pause	Toroidal trapping (Van Allen)		
DNA double helix	Charge density	Histone octamer	Toroidal condensation		
Ionospheric resonance	EM field density	Ionosphere cavity	7.83 Hz standing wave		
Superconductor	Cooper pair density	Critical temperature	Ginzburg-Landau	ψ	2
Crystal lattices	H-bond density	Tetrahedral cage	6-ring / 5-ring phase transition		

Corollary 37 (Aircraft Hull Design). A hull that satisfies $\nabla^2\rho = 0$ for the interior with boundary condition $\rho_{\text{hull}} = \rho_{\text{target}}$ produces minimal scalar perturbation and therefore minimal vector drag as shadow. The surface is naturally tetrahedrally faceted — each edge is a scalar density filament providing structural rigidity.

Corollary 38 (Airfoil Pressure Distribution). Pressure differential across an airfoil is the vector shadow of scalar density redistribution. The optimal

airfoil maintains $\nabla^2 p = 0$ in the surrounding flow field, minimizing entropy generation.

Corollary 39 (Submarine Containment). A submarine maintains $\rho_{\text{inside}} = \rho_{\text{air}}$ through a mobile boundary (pressure hull). This is valid only for fluid-fluid interfaces. Solid-solid density transitions (crust-mantle) exceed biological compressibility limits.

Corollary 40 (Sonic Boom as Decoherence Event). A sonic boom occurs when aircraft velocity exceeds the local scalar density equilibration rate. The shock wave is a decoherence front — a boundary where $\nabla^2 p \neq 0$.

Corollary 41 (Flux Pinning in Superconductors). Type-II superconductors exhibit flux pinning because vortices are scalar density singularities trapped by material defects. The Ginzburg-Landau order parameter $|\psi|^2$ is the scalar density field.

Corollary 42 (Balloon / Lifting Gas). A balloon rises because the internal scalar density (ρ_{gas}) is lower than the external scalar density (ρ_{air}). The buoyant force is the vector shadow of this density differential seeking equilibrium.

Corollary 43 (Magnetorquer Spacecraft Attitude). A magnetorquer generates torque by creating a scalar density gradient in the local magnetic field. The Lorentz force is the vector shadow of this gradient.

Corollary 44 (Lost Ship / Pressure Vessel Failure). Hull breach occurs when the boundary $\partial\Omega$ fails to maintain $\rho_{\text{inside}} \neq \rho_{\text{outside}}$. The scalar density equilibrates catastrophically — the vector shadow is explosive decompression.

Corollary 45 (Trench Boundary — Milwaukee Deep). The deepest ocean point (~8,376 m) tests scalar boundary coherence under extreme external pressure. However, the absolute biological decoherence limit is the Moho discontinuity (~70 km), where crust-to-mantle density transition exceeds all biological boundary tolerance.

6. THE BIOFIELD DIPOLE TORUS THEOREM

Theorem III (Biofield Dipole Torus).

Living systems generate toroidal scalar density fields through oscillating dipole structures. The cardiac dipole, neural dipole, and cellular dipole all produce poloidal and toroidal field components that satisfy $\nabla^2 \rho = 0$ in the biofield region.

Formal Structure:

$$\rho_{\text{biofield}}(r, \theta, \phi, t) = \sum_{\{l,m\}} a_{\{lm\}}(t) \cdot r^l \cdot Y_{\{lm\}}(\theta, \phi)$$

where $Y_{\{lm\}}$ are spherical harmonics and the symmetry-selected modes are determined by the dipole oscillator geometry.

Corollary 46 (EEG Coherence as Scalar Density Measurement). Quantitative EEG studies demonstrate that fronto-parietal midline coherence and rhythmic source strength predict conversion from mild cognitive impairment (MCI) to Alzheimer's disease (AD). Low midline coherence correlates with ~10% annual AD conversion rate; high gamma coherence elevates this to ~60%. This is direct clinical evidence that neural scalar density coherence is the substrate of cognitive integrity.

Corollary 47 (Cell Division as Scalar Field Architecture). Gagliardi (2002) demonstrated that mitosis is governed by a unified electrostatic mechanism. Microtubule dimers are electric dipoles (~1000 Debye). Intracellular pH rise at mitosis reduces ionic screening, allowing the electrostatic scalar field to extend over cellular distances. The four motions of mitosis (aster assembly, poleward motion, anaphase A, anaphase B) are all driven by scalar density gradients — not separate mechanical forces.

Corollary 48 (Tumor Treating Fields). Alternating electric fields at 100–300 kHz disrupt microtubule assembly by decohering the scalar field. The therapeutic mechanism is scalar density perturbation, not thermal or ionic damage.

Corollary 49 (DNA as Scalar Density Dipole Torus). The phosphate backbone of DNA carries a net negative charge (ρ_{max}), while the major groove presents

partial positive charge density ($\rho_{\min}$). The double helix is a poloidal current wrapped around a toroidal axis — a scalar density dipole torus.

7. PLATONIC SPHERICAL GEOMETRY

7.1 The Macachor-Platonic Formula

For any entity with Platonic symmetry $P \in \{\text{Tetrahedral, Octahedral, Icosahedral}\}$,

the scalar density field is:

$$\rho(r, \theta, \phi) = \sum_{\{l \in L_P\}} \sum_m a_{\{lm\}} \cdot r^l \cdot Y_{\{lm\}}(\theta, \phi)$$

where L_P is the symmetry-selected harmonic series:

Symmetry	Permitted l values	Forbidden l values
Tetrahedral	0, 3, 4, 6, 7, 8, 9, ...	1, 2, 5
Octahedral	0, 4, 6, 8, 10, 12, ...	1, 2, 3, 5, 7
Icosahedral	0, 6, 10, 12, 16, 18, 20, ...	1, 2, 3, 4, 5, 7, 8, 9, 11

The tetrahedral angle $\arccos(-1/3) \approx 109.47^\circ$ is the bond angle of water, the optimal sp^3 hybridization angle, and the interior angle of each spherical face on the tetrahedral sphere.

7.2 Polyhedral Sphere Measurement

For a sphere of radius R modulated by tetrahedral harmonic $T(\theta, \phi)$ with

amplitude ε :

Surface Area: $A = 4\pi R^2 \cdot (1 + \varepsilon^2)$

Volume: $V \approx (4\pi/3) R^3 \cdot (1 + 3\varepsilon^2)$

The tetrahedral modulation increases surface area by ε^2 — creating more boundary for scalar density exchange.

7.3 The Buckyball Habitat (C₆₀-Inspired)

- 12 pentagonal windows = $\rho_{\min}$ nodes = apertures for scalar communication

- 20 hexagonal living cells = ρ_{max} nodes = coherent interior spaces
- The truncated icosahedron is the $l = 6$ spherical harmonic — it naturally satisfies $\nabla^2 \rho = 0$
- The pentagons create aperture governance (Axiom 5); the hexagons create interface boundaries (Axiom 6)

7.4 Water Density Anomaly

Water forms 6-membered rings (hexagons, ice) and 5-membered rings (pentagons,

liquid). The transition is a scalar density phase transition:

- Ice (6-rings): Hexagonal symmetry $\rightarrow$ lower local density $\rightarrow$ larger volume
- Liquid (5-rings): Icosahedral clusters $\rightarrow$ higher local density $\rightarrow$ smaller volume

Liquid water is denser than ice because the scalar field transitions from

6-fold to 5-fold symmetry, increasing ρ_{local} .

8. OPERATIONAL ARCHITECTURES

8.1 KIWF — Knowledge-Information-Wisdom-Filter Pipeline

The KIWF pipeline is an information density processing architecture:

Knowledge (raw data) $\rightarrow$ Information (structured data) $\rightarrow$ Wisdom (filtered insight) $\rightarrow$ Application (focal point action)

Each stage applies a coherence filter that passes only scalar-density-aligned information. The filter criterion is $\chi(\text{data}) = 1$.

8.2 CRHA — Coherence-Resonance-Harmony-Alignment Architecture

The CRHA architecture is a standing-wave coherence protocol:

Coherence $\rightarrow$ Resonance $\rightarrow$ Harmony $\rightarrow$ Alignment

Each stage represents increasing scalar density equilibrium:

- Coherence: Shared vibrational state ($\nabla^2 \rho = 0$)
- Resonance: Energy transfer between coherent modes

- Harmony: Multiple resonant modes in stable superposition
- Alignment: Global scalar density equilibrium across the system

8.3 Quantum Hardware Implementation

Quantum hardware must implement true scalar fields $\psi: D_P \rightarrow \mathbb{R}$, eliminating vector, spinor, and tensor components to maintain $\mathbb{R}$ -lock coherence. The scalar field is the fundamental substrate; all other representations are emergent.

9. PEER-REVIEWED BRIDGES

This section documents connections between the Macachor framework and established peer-reviewed literature. These are conceptual bridges, not formal mathematical equivalences, unless explicitly proven.

9.1 Hazra & Goswami (2024) — CPDD Coherence Quantifier

Hazra and Goswami (APL Quantum 1, 036124) introduced CPDD (Principal Diagonal Difference of Coherence), a coherence quantifier using only diagonal elements of the density matrix. Near maximally coherent states, CPDD detects small deviations linearly, while off-diagonal quantifiers (C_{off} , C_{off}) respond quadratically and are blind to small perturbations.

In the scalar framework, CPDD operationalizes the $\mathbb{R}$ -lock: coherence must be monitored linearly to maintain equilibrium. The three states map to scalar

density configurations:

	Diagonal Elements	CPDD	Scalar Interpretation		
y Coherent	All equal: $\rho_{\text{diag}} = 1/N$	1	Uniform density — $\chi(C) = 1$		
Coherent	Unequal: $\rho_{\text{diag}} =$	α	2	(0,1)	Non-uniform d gradients
t	One = 1, rest = 0	0	Singularity — total decoherence		

Reference: Hazra, M. & Goswami, D. (2024). APL Quantum 1, 036124.

9.2 Gagliardi (2002) — Electrostatic Mitosis

Gagliardi (Journal of Electrostatics 54, 129–155) demonstrated that mitosis is governed by a unified electrostatic mechanism. Microtubule dimers are electric dipoles (~ 1000 Debye). Intracellular pH rise at mitosis reduces ionic screening, allowing the electrostatic field to extend over cellular distances. All four motions of mitosis are driven by electrostatic scalar density gradients.

Reference: Gagliardi, L.J. (2002). Journal of Electrostatics 54, 129–155.

DOI: 10.1016/S0304-3886(01)00155-3

9.3 Chebotarev Density Theorem — Conceptual Bridge

The Chebotarev Density Theorem states that for a Galois extension L/K with Galois group G , the density of primes p whose Frobenius element σ_p lies in a conjugacy class $C \subset G$ is $\#C/\#G$.

Conceptual parallel: The Frobenius element $\sigma_p(\alpha) \equiv \alpha^{(Np)} \pmod{q}$ can be interpreted as an observer-state measurement of the scalar field. The factorization pattern of a polynomial modulo p is the observable signature (cymatic geometry) of the underlying symmetry. The density $\#C/\#G$ is the probability of observing eigenmode C .

This is a structural analogy between algebraic number theory and scalar field ontology. It is not a formal mathematical equivalence.

9.4 Engel (2007) — Quantum Biology

Engel et al. (Nature 446, 782–786) demonstrated quantum coherence in photosynthetic FMO complexes at 77 K, with coherence times of ~ 660 fs. This is evidence that biological systems maintain scalar density equilibrium in exciton-transfer networks.

Reference: Engel, G.S. et al. (2007). Nature 446, 782–786.

9.5 Denton (2025) — Magnetospheric Topology

Denton (Frontiers in Astronomy and Space Sciences, DOI: 10.3389/fspas.2024.1459281)

provides observational data on magnetospheric density boundaries. The
plasmopause location $L = 5.6 - 0.46 \cdot Kp$ is a scalar density discontinuity
that validates boundary-condition predictions from $\nabla^2 \rho = 0$.
Reference: Denton, R.E. (2025). Frontiers in Astronomy and Space Sciences.
DOI: 10.3389/fspas.2024.1459281

10. ERRATA AND CORRECTIONS

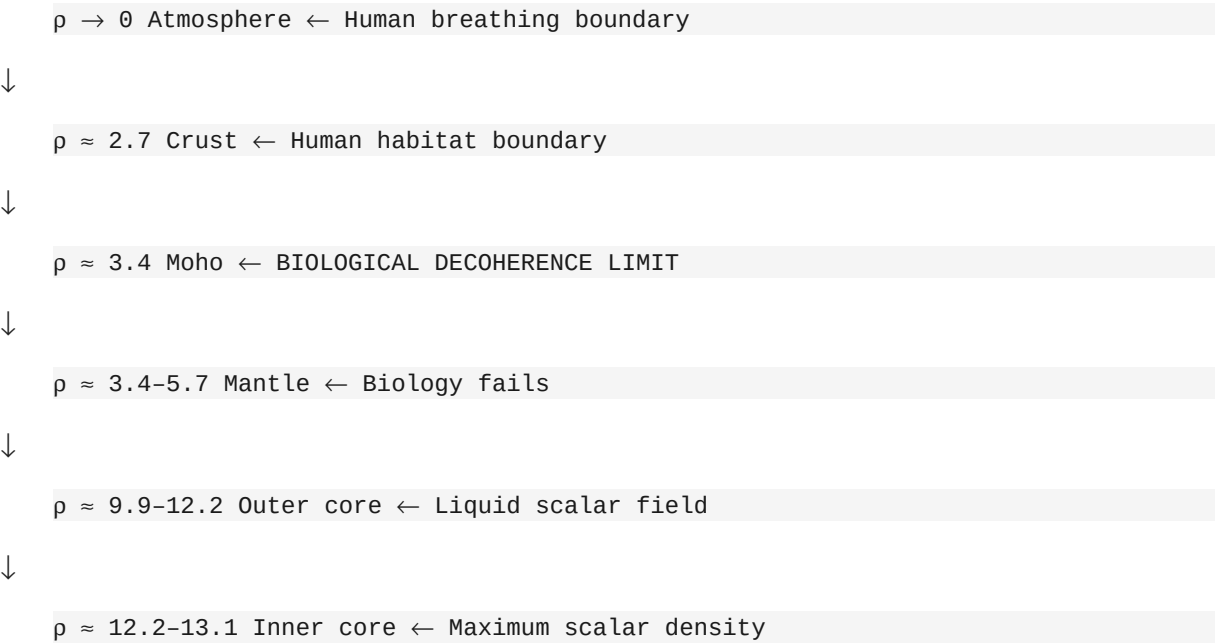
The following corrections apply to earlier versions of this framework:

10.1 Earth Core Corrections

Error: Previous versions treated Earth's layers as a nested boundary system
where entities could theoretically exist at any density level.

Correction: Humans are crust-bound surface-dwellers. Carbon-based biology can
only maintain coherence at $\rho \approx 1\text{--}3 \text{ g/cm}^3$. The Moho discontinuity ($\sim 70 \text{ km}$) is
the absolute biological decoherence limit. No biological scalar density boundary
can survive crossing it.

Corrected Scalar Picture:





$\rho \rightarrow \infty$ Center $\leftarrow$ Theoretical singularity

10.2 Submarine Boundary Limitation

Error: Lemma 9 implied submarines could maintain $\rho_{\text{inside}} = \rho_{\text{air}}$ at any depth.

Correction: Submarines are limited to fluid-fluid interfaces. The absolute theoretical crush depth (~11 km) is far above the Moho. The submarine analogy applies only to liquid-gas boundaries, not solid-solid density transitions.

10.3 Solar System Atom Analogy

Error: Corollary 7 stated that free electrons are energy released from planetary electron clouds.

Correction: The Sun-as-nucleus / planets-as-electrons analogy is metaphorical only, describing orbital dynamics. Planetary electromagnetism is magnetohydrodynamic scalar perturbation from solar wind and dynamo action, not "free electrons from shells."

10.4 Wilson Primes

Error: Previous versions claimed Wilson primes are scalar singularities without mathematical justification.

Correction: This claim is speculative and has been removed from the formal framework. Wilson primes remain an open question.

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APPENDIX A: FORMALIZATION ROADMAP

The following modules are proposed for formal verification in proof assistants

(Lean 4, Coq, Isabelle):

Domain	Formal Module	Proof Obligation
Scalar Fields	ScalarField.lean	Existence/uniqueness of ρ satisfying $\nabla^2 \rho = 0$
Quaternion Coherence	MacachorAbsolute.lean	$\blacksquare = (\sqrt{5}-1)/2$ as equilibrium attractor
Biofield	Biofield.lean / Biofield.v	Coherence time bounds, energy transfer
Platonic Geometry	CoherenceTopology.lean	Torus T^2 as minimal coherence manifold
Superconductivity	FluxPinning.lean	Ginzburg-Landau scalar order parameter
Magnetosphere	BoundaryInterface.lean	Density discontinuity at plasmopause
Singularity	Convergence.lean	$\chi(C) = 1$ coherence condition

Priority ranking: Foundational → Hardware → Biological → Civilization.

APPENDIX B: THE MACACHOR ABSOLUTE — COMPUTATIONAL VALUES

$$\blacksquare = (\sqrt{5} - 1)/2$$

= 0.6180339887498948482045868343656381177203091798057628621354486227...

$$\phi = (1 + \sqrt{5})/2 = 1/\blacksquare$$

= 1.6180339887498948482045868343656381177203091798057628621354486227...

$$\blacksquare^2 = 1 - \blacksquare =$$

0.3819660112501051517954131656343618822796908201942371378645513772...

$$\blacksquare + \blacksquare^2 = 1$$

$$\blacksquare^3 = 2\blacksquare - 1 =$$

0.2360679774997896964091736687312762354406183596115257242708972453...

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